

# CE 4011 Static + Modal Desktop MVP Verification

## Appendix

Final Documentation Package

Spring Semester 2025–2026

### 1 Purpose and Scope Boundary

This appendix records detailed validation evidence for the submitted desktop MVP. The final desktop analysis scope is Static Analysis and Modal Analysis only. RSA and THA may remain in the repository as deferred/future-extension backend work, but they are not part of the submitted desktop verification claim.

The main report summarizes the validation baseline. This appendix provides the longer benchmark descriptions, acceptance criteria, coverage mapping, and comparison tables. Values from the existing `data/` result exports are inserted where the exported quantity is explicit. Entries remain TODO only where the required developed-software value, reference value, percent difference, or pass/fail decision is not available in the retained evidence files.

### 2 Verification Method

The verification strategy combines reference benchmarks and automated tests. Static benchmark models are loaded from XML, solved through the same DOF/assembly/solver/post-processing path used by the desktop application, and compared with SAP2000 text-output references. Modal benchmark models are loaded from XML, assembled dynamically, reduced or condensed where needed, and compared with a flexible-beam modal reference solution.

Table 1: Verification evidence map

| Evidence area          | Input/reference                                | Quantities checked                                                                                             | Acceptance basis                                                                                                    |
|------------------------|------------------------------------------------|----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| Static frame benchmark | Frame XML; SAP2000 frame text output           | Nodal displacements, support reactions, member-end forces, global/reduced matrices retained for inspection     | Regression comparison against parsed SAP2000 tables, with relative tolerances used by the automated benchmark tests |
| Settlement benchmark   | Settlement XML; SAP2000 settlement text output | Prescribed support settlement response, nodal displacements, reactions, member forces, SAP2000-display mapping | SAP2000 comparison plus display-mapping tolerance used by desktop/static tests                                      |

| Evidence area                 | Input/reference                                      | Quantities checked                                                                                                             | Acceptance basis                                                                   |
|-------------------------------|------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| Temperature benchmark         | Temperature XML; SAP2000 temperature text output     | Thermal fixed-end effects, inclined truss transformations, displacements, reactions, member-end forces                         | Strict thermal benchmark tolerances for displacement, force, and moment components |
| Modal flexible-beam benchmark | Modal XML; flexible-beam modal reference text output | Condensed stiffness, eigenvalues, circular frequencies, periods, mode shapes, mass-normalized modal properties where available | Absolute tolerances used by modal regression tests                                 |
| Desktop command paths         | Headless desktop test doubles and temporary files    | XML save/export, XML parse-back, Static table TXT export, Modal table CSV export, Static plot PNG export                       | Final P0 desktop workflow tests                                                    |

The retained benchmark inputs are `data/test-frame.xml`, `data/test-settlement.xml`, `data/test-temperature.xml`, and `data/test-modal-flex-beam.xml`. The corresponding reference outputs are stored under `sap2000_solutions/` and `modal_solution/`. The final documentation export evidence files are `data/test_frame_static_results_table.txt` and `data/modal_results_table.csv`.

### 3 Detailed Theory Notes

This section retains the longer equations and implementation notes removed from the main report to keep the submitted main body within the CE 4011 page limit.

For a two-dimensional frame member with local axial and flexural behavior, the local stiffness contribution has the standard direct-stiffness form

$$k_\ell = \begin{bmatrix} EA/L & 0 & 0 & -EA/L & 0 & 0 \\ 0 & 12EI/L^3 & 6EI/L^2 & 0 & -12EI/L^3 & 6EI/L^2 \\ 0 & 6EI/L^2 & 4EI/L & 0 & -6EI/L^2 & 2EI/L \\ -EA/L & 0 & 0 & EA/L & 0 & 0 \\ 0 & -12EI/L^3 & -6EI/L^2 & 0 & 12EI/L^3 & -6EI/L^2 \\ 0 & 6EI/L^2 & 2EI/L & 0 & -6EI/L^2 & 4EI/L \end{bmatrix}.$$

Truss members use the axial subspace of this relationship. Each element contribution is transformed into global coordinates, assembled into the global stiffness matrix, and partitioned into free and restrained degrees of freedom. With restrained displacement vector  $u_r$ , the solved static system is

$$K_{ff}u_f = F_f - K_{fr}u_r, \quad R = Ku - F.$$

Element end forces are recovered by returning the solved global displacement vector to local member coordinates and adding equivalent nodal-load effects where member loads, settlement, or temperature loads apply. The same recovered local-force data supports axial-force, shear-force, and bending-moment diagram sampling.

The modal workflow solves the generalized eigenproblem

$$K_d \phi_i = \lambda_i M_d \phi_i, \quad \omega_i = \sqrt{\lambda_i}, \quad T_i = \frac{2\pi}{\omega_i},$$

on the active dynamic degrees of freedom. Massless DOFs that are stiffness-coupled to retained inertial DOFs are statically condensed rather than deleted:

$$K_c = K_{aa} - K_{am} K_{mm}^{-1} K_{ma}.$$

Direct deletion is reserved for disconnected zero-mass DOFs. Mode shapes are normalized for display and for modal-property reporting. Modal mass, participation factor, effective modal mass, and participation ratio are computed from the retained mass matrix and the selected excitation direction.

## 4 Desktop Screenshot Evidence

The main report keeps one UI/workflow figure and one result/visualization figure. The remaining screenshots are retained here as detailed evidence for the installation/user workflow, model setup, validation feedback, static outputs, modal outputs, and visualization/export behavior.

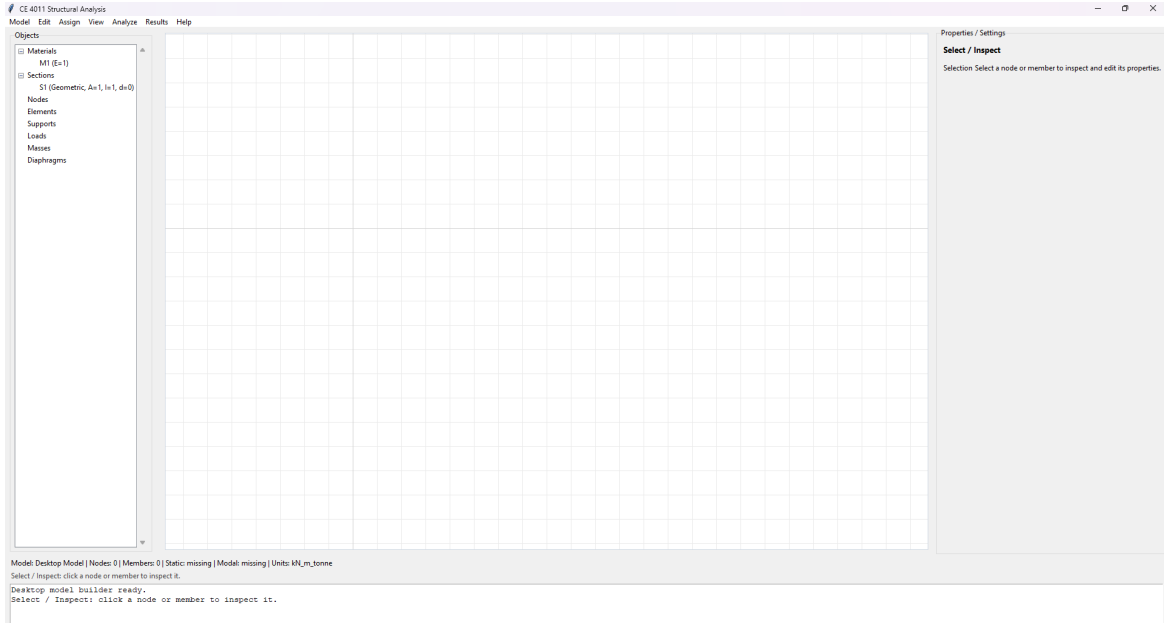


Figure 1: Main desktop UI

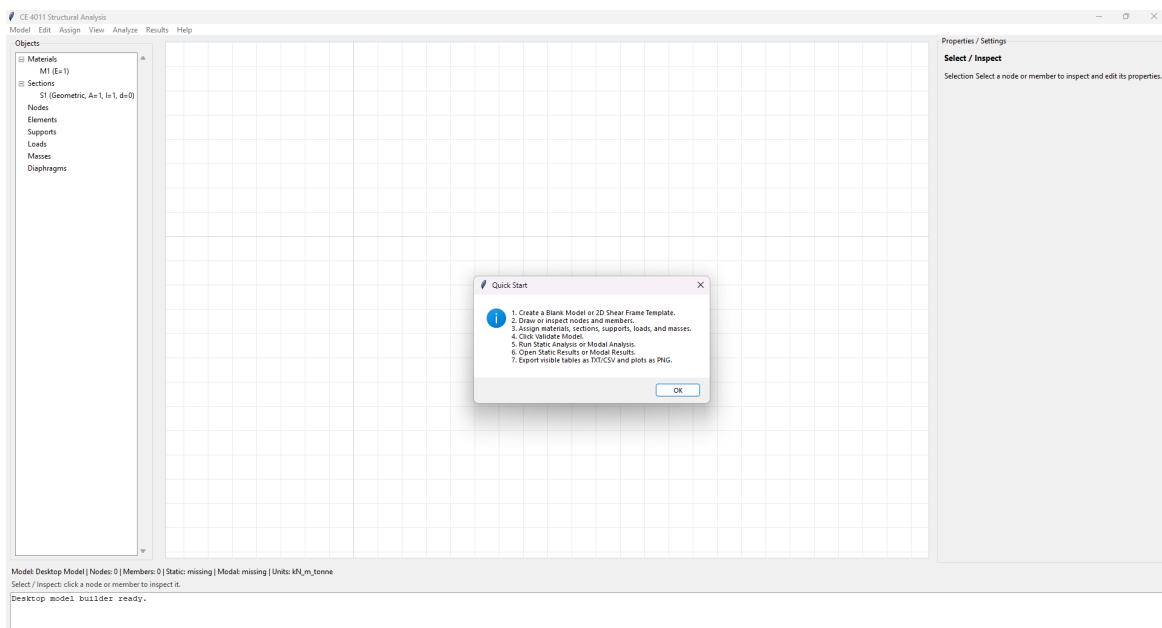


Figure 2: Quick-start workflow screen

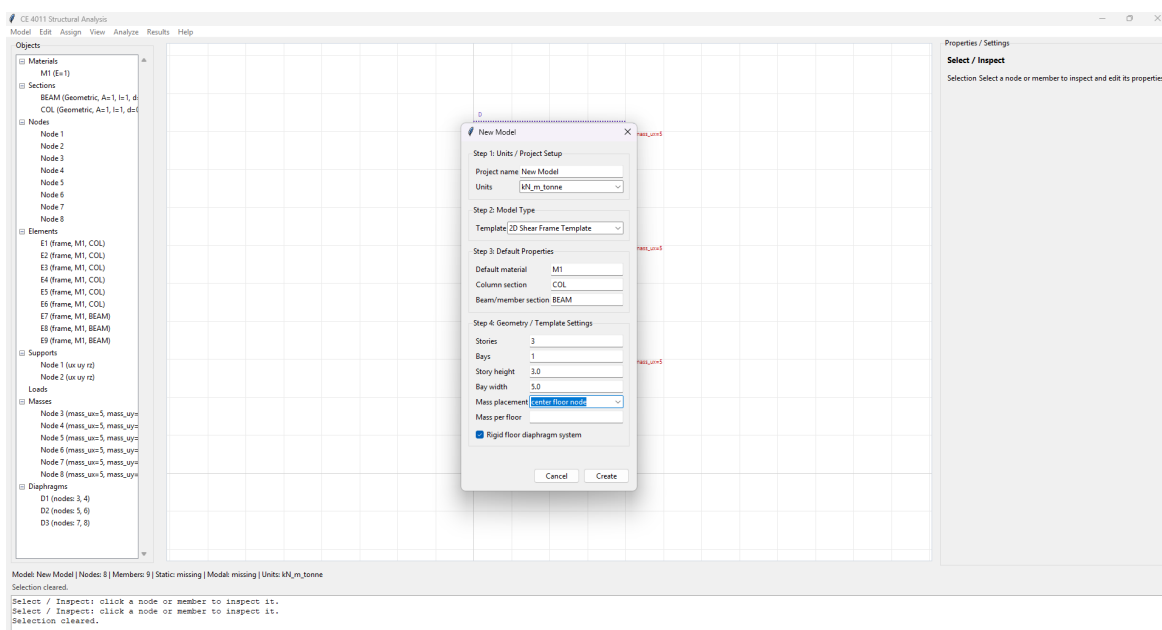


Figure 3: Shear-frame template input

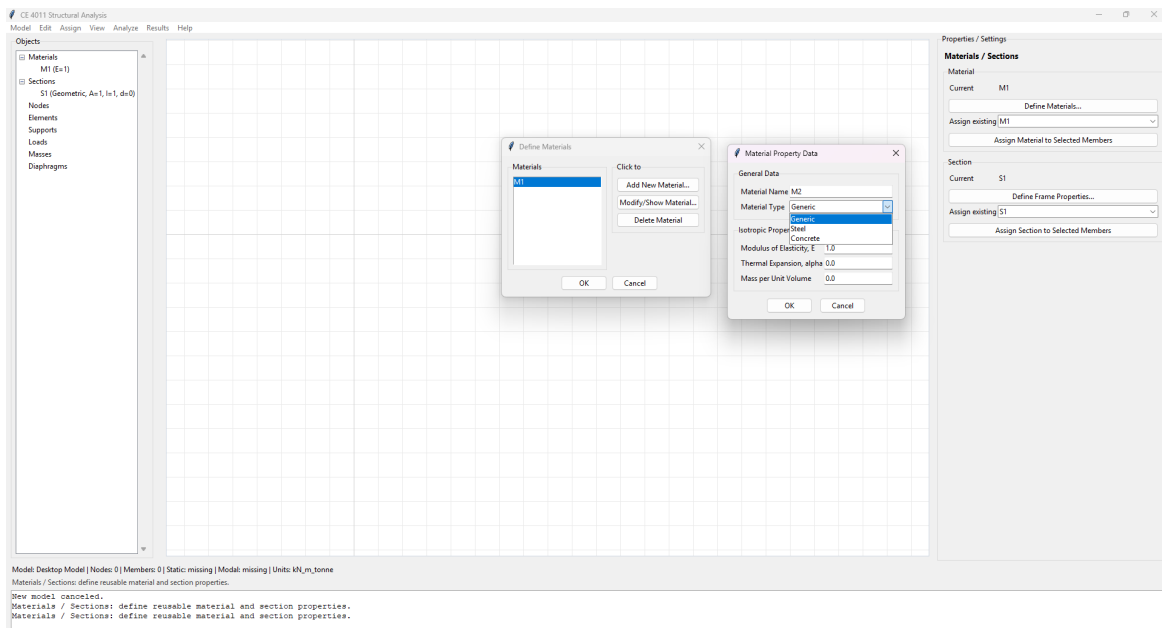


Figure 4: Property panel

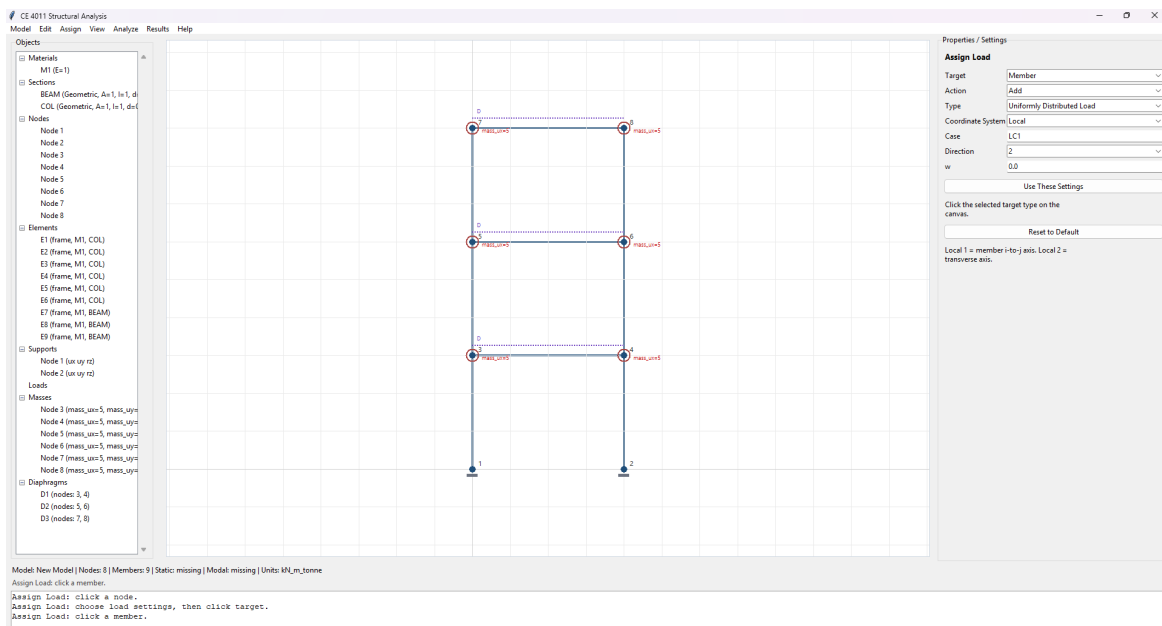


Figure 5: Member load assignment

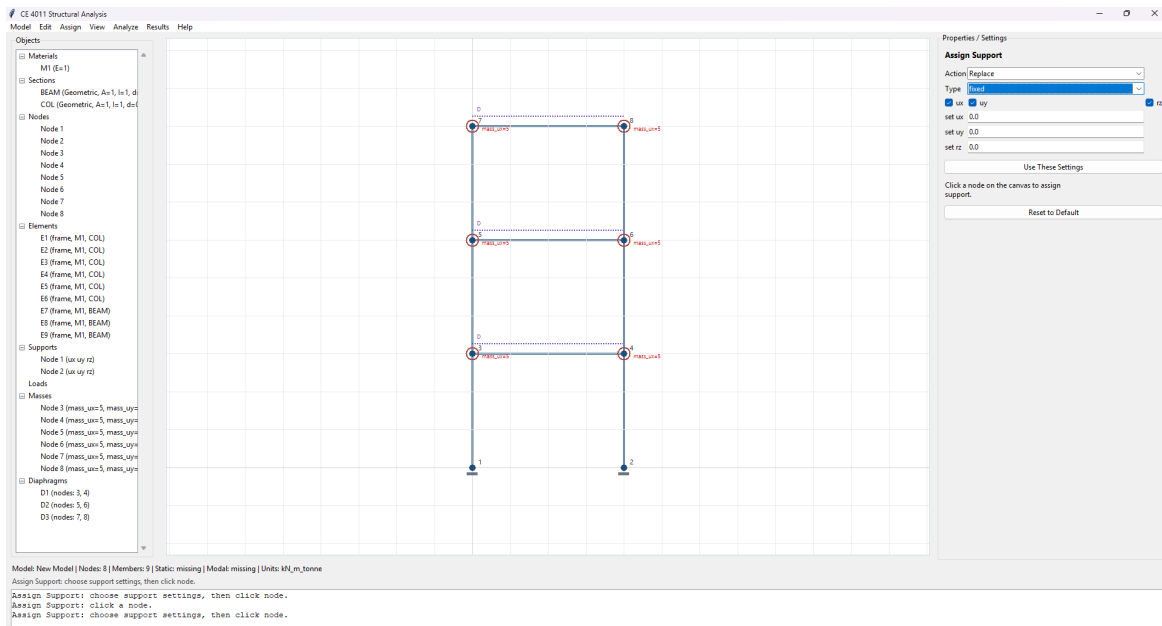


Figure 6: Support assignment

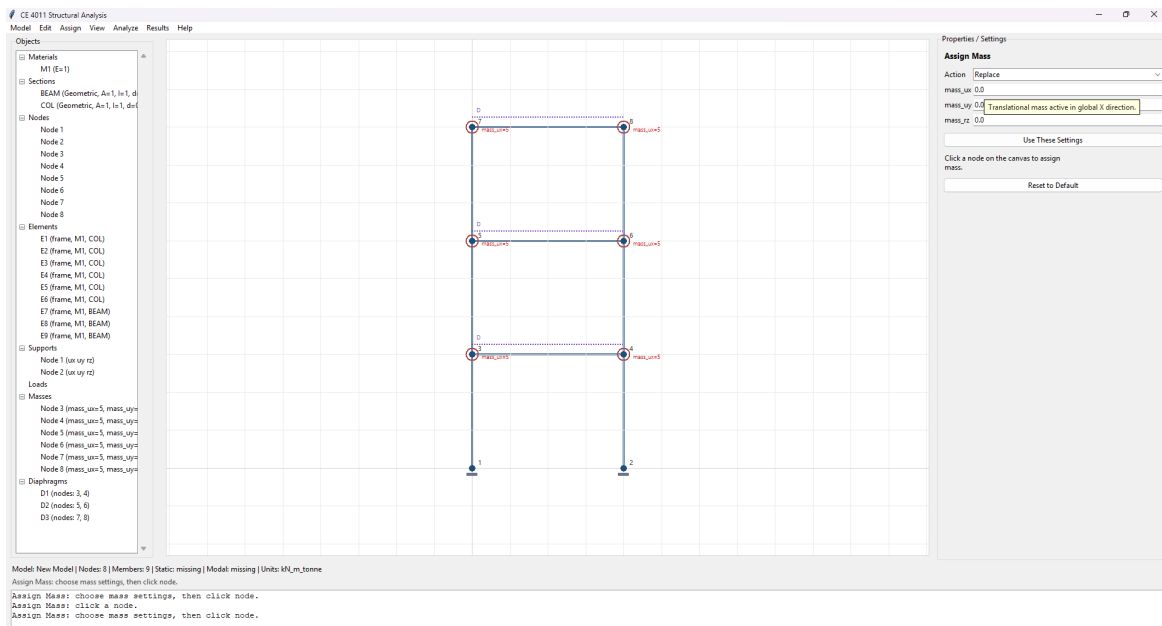


Figure 7: Mass assignment

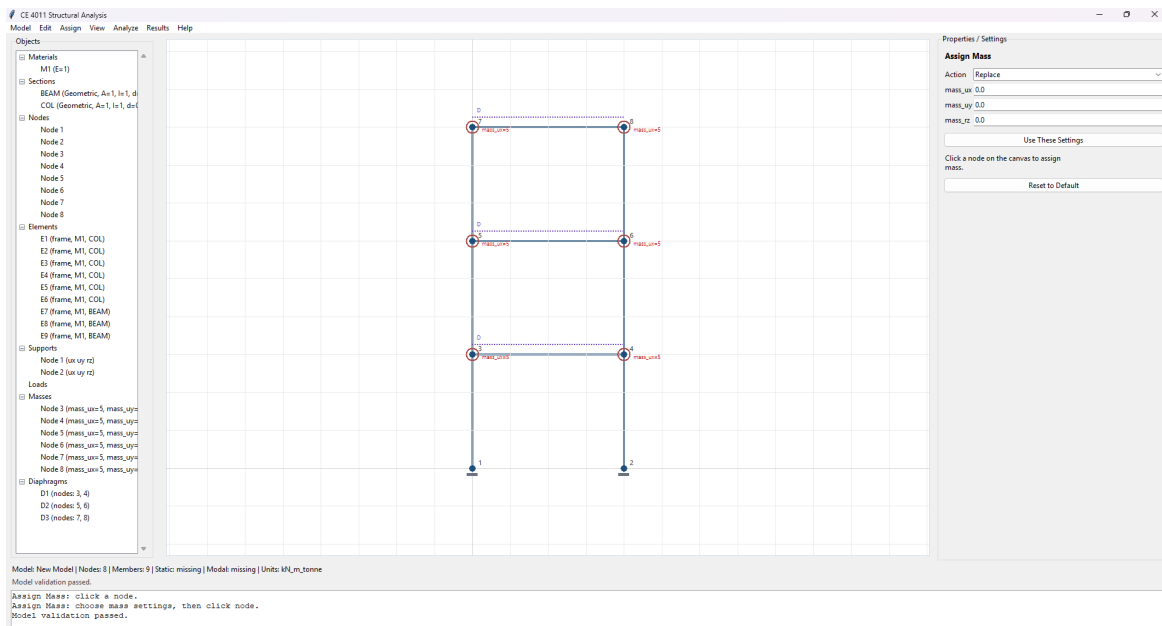


Figure 8: Successful model validation

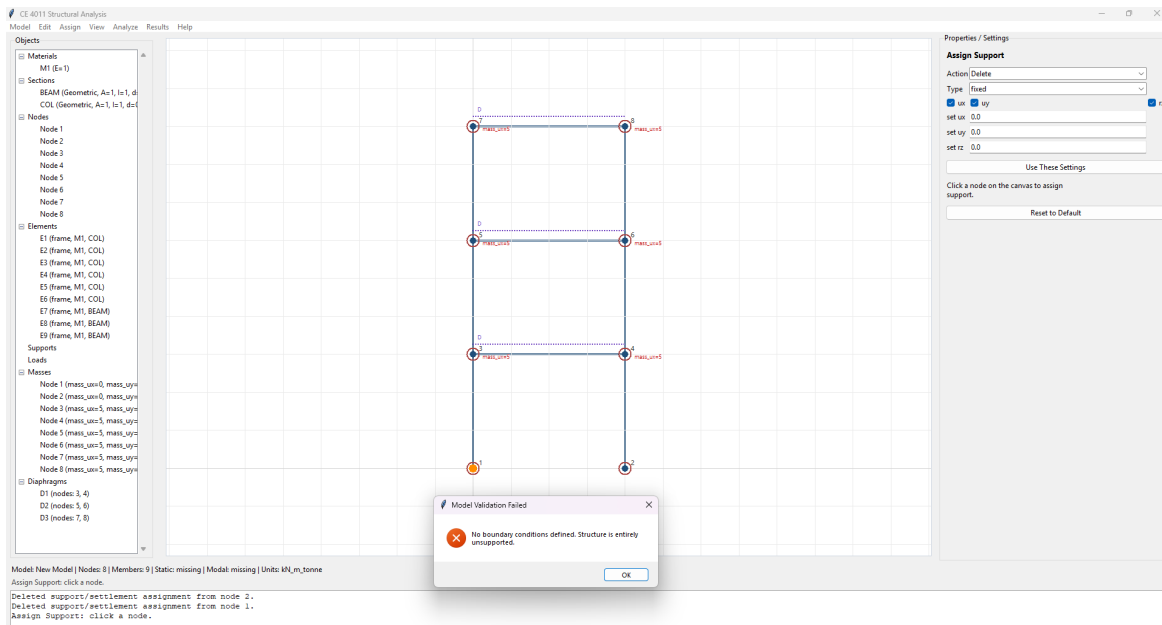


Figure 9: Unsuccessful model validation

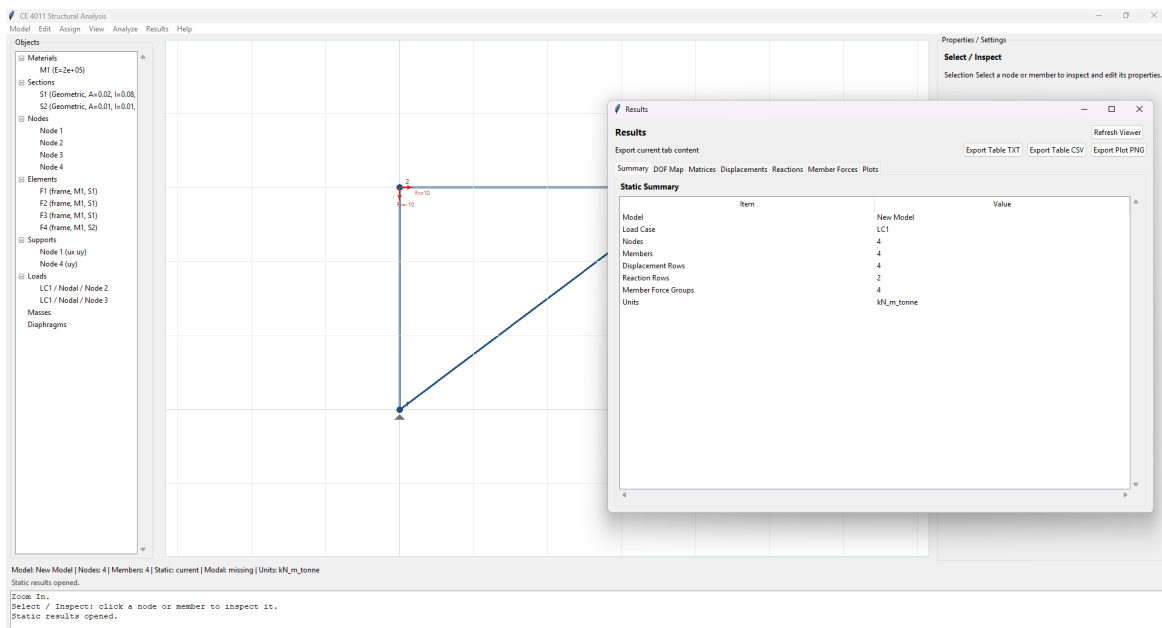


Figure 10: Static result summary

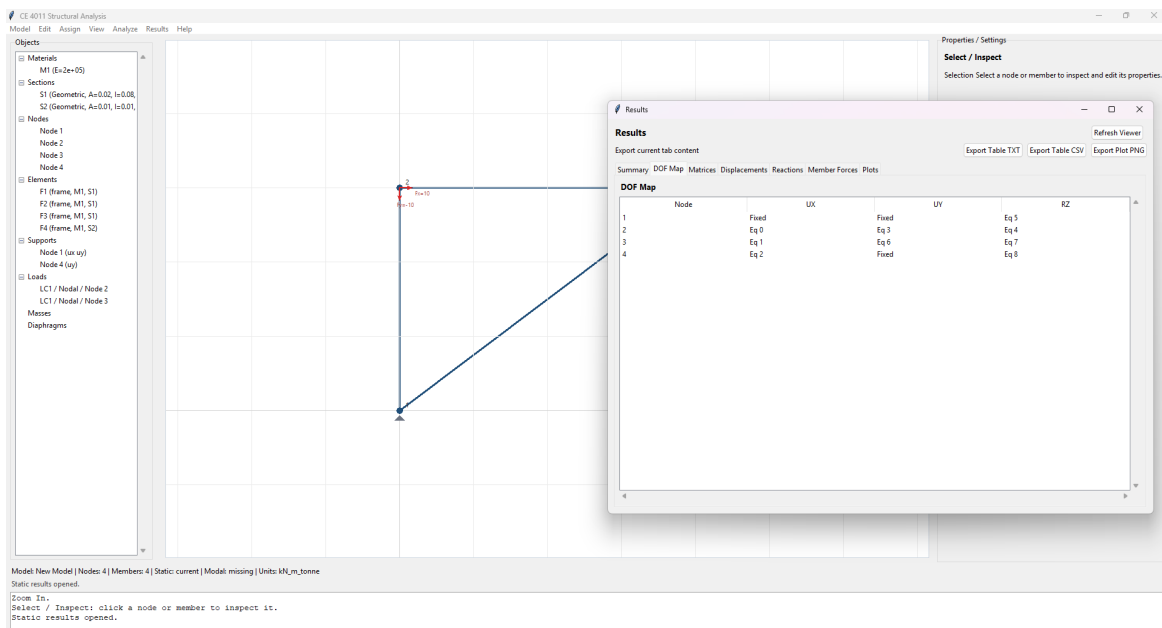


Figure 11: Static DOF map



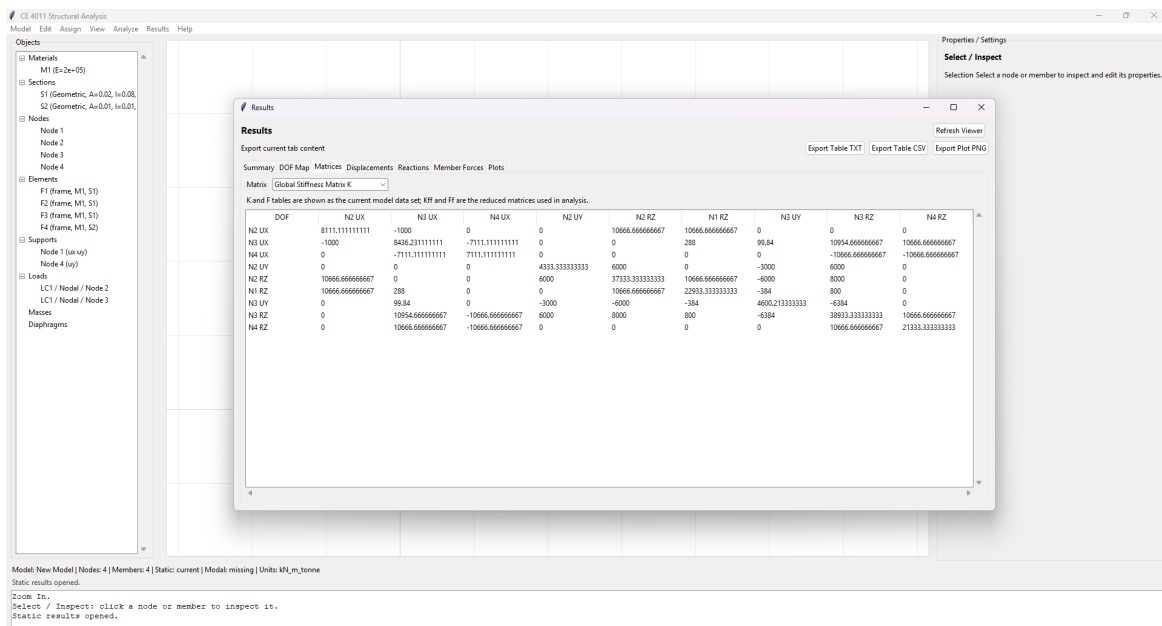


Figure 12: Static stiffness matrix

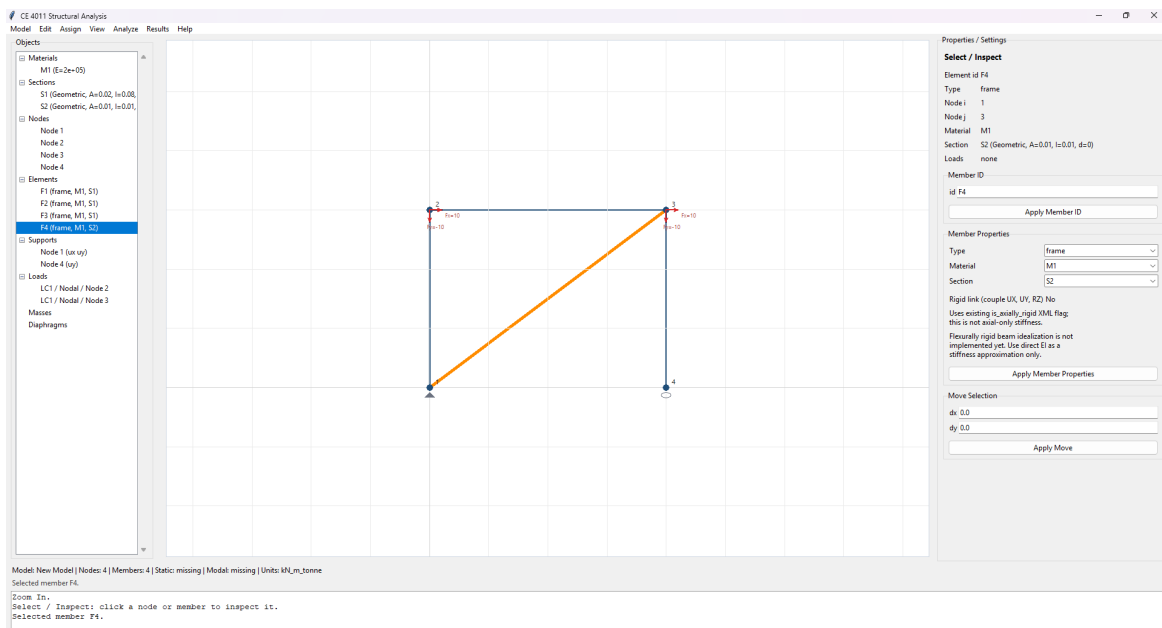


Figure 13: Static test-frame model

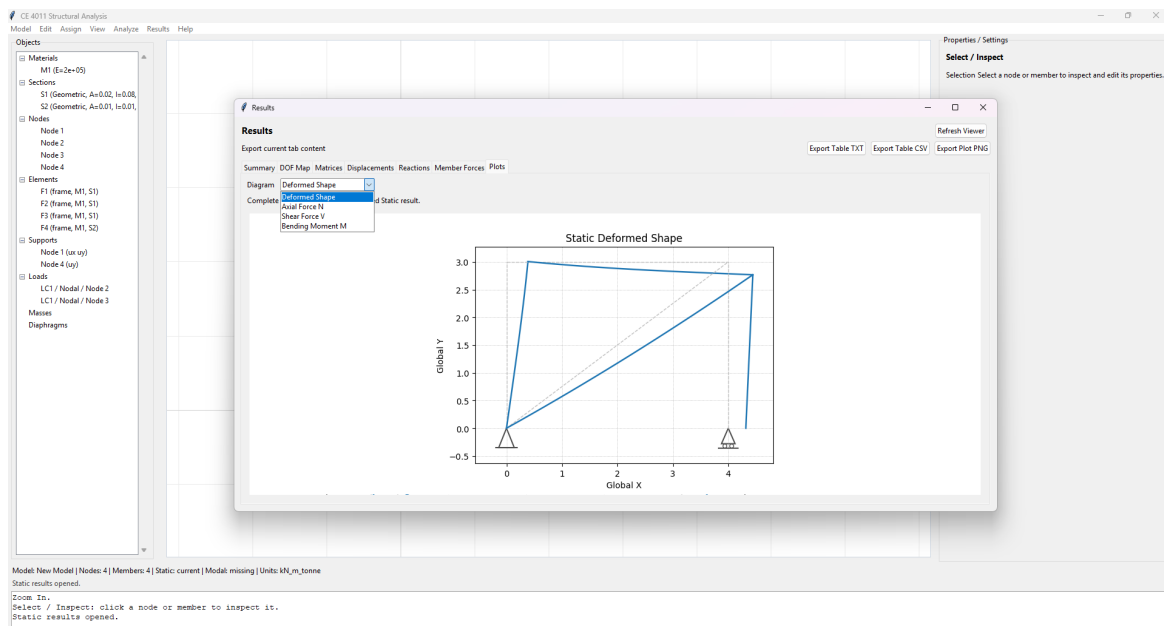


Figure 14: Static deformed shape

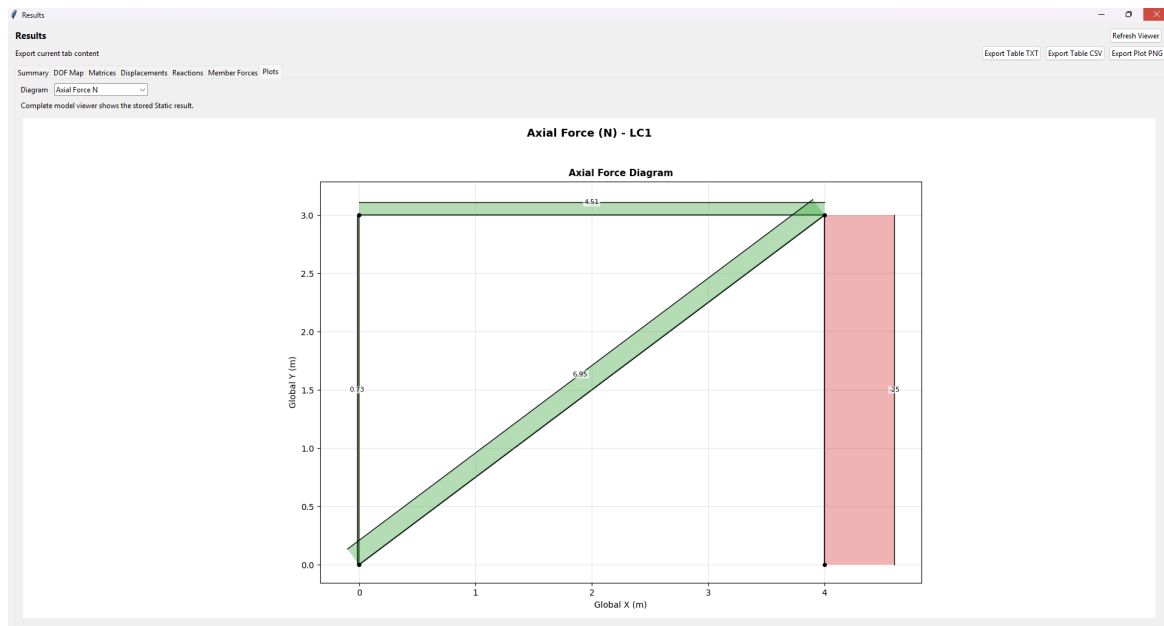


Figure 15: Static axial-force diagram

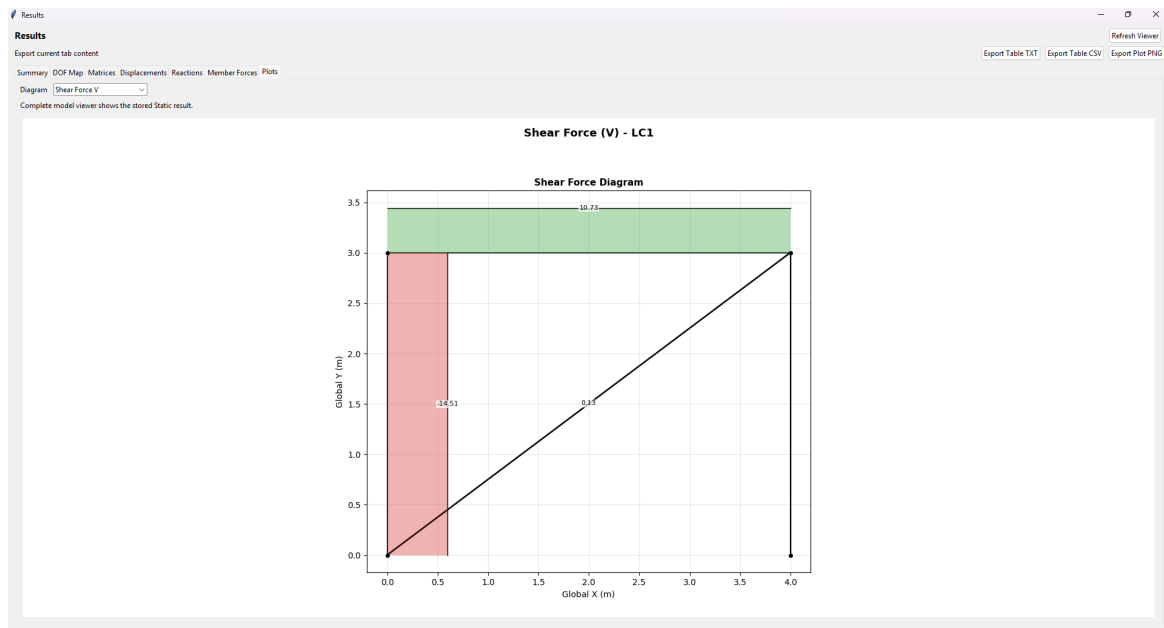


Figure 16: Static shear-force diagram

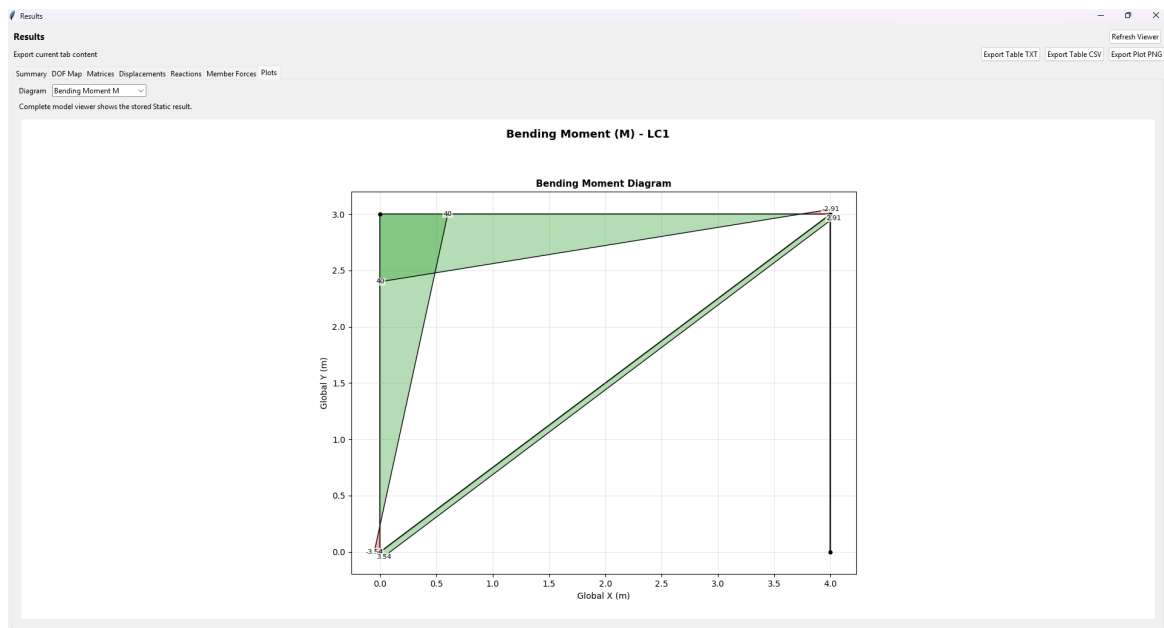


Figure 17: Static bending-moment diagram

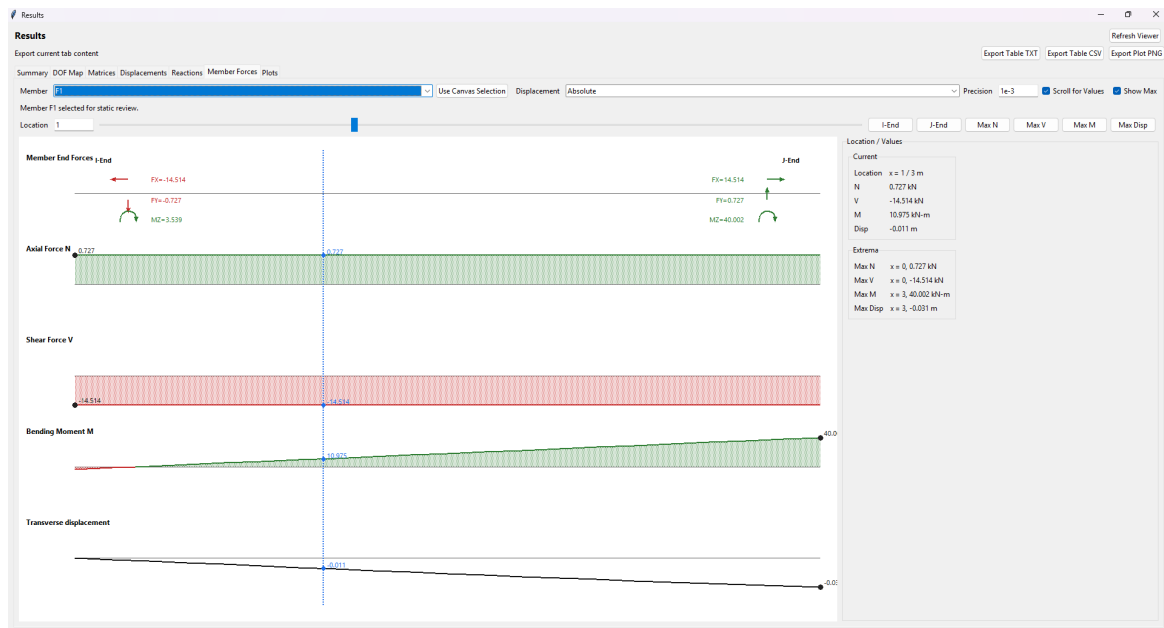


Figure 18: Individual member static result

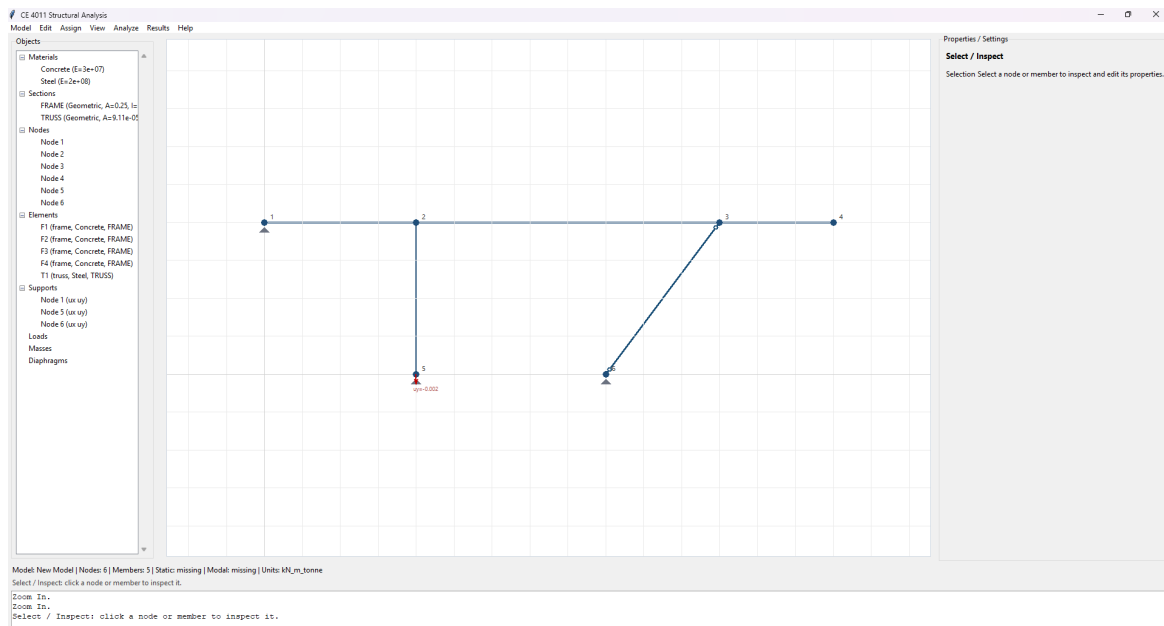


Figure 19: Settlement case screen

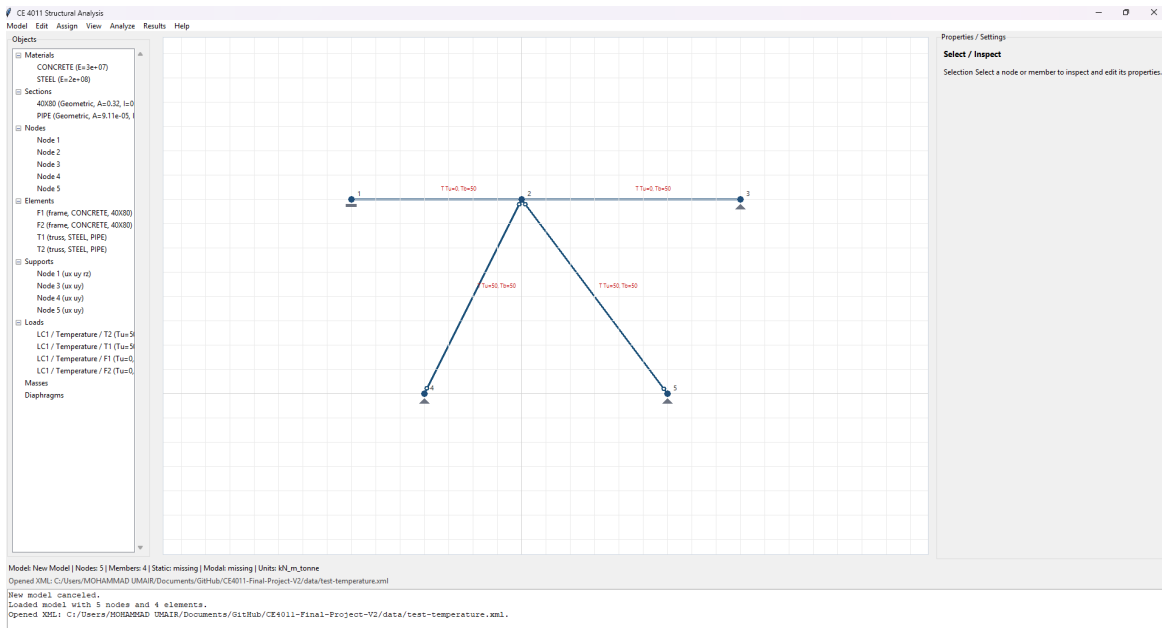


Figure 20: Temperature case screen

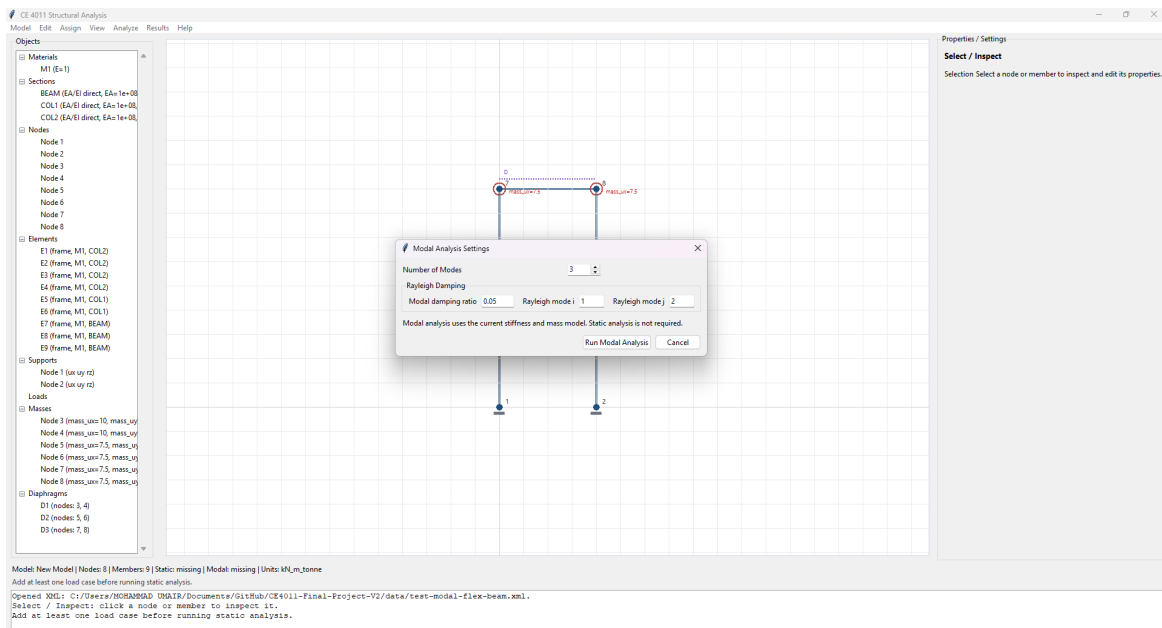


Figure 21: Modal settings

Results

Results

Export current tab content

Summary DOF Map Matrices Modal Table Mode Shapes

Precision 1e-3 Apply

Modal Summary

| Mode | Eigvalue ( $\omega^2$ ) | Omega ( $\omega$ ) | Frequency | Period | Modal Mass | Participation Factor | Effective Modal Mass | Mass Participation Ratio | Modal Damping Ratio |
|------|-------------------------|--------------------|-----------|--------|------------|----------------------|----------------------|--------------------------|---------------------|
| 1    | 20.737                  | 4.554              | 0.725     | 1.38   | 1          | 6.296                | 39.637               | 79.274%                  | 5%                  |
| 2    | 209.652                 | 14.479             | 2.304     | 0.434  | 1          | 2.821                | 7.958                | 15.917%                  | 5%                  |
| 3    | 807.042                 | 28.408             | 4.521     | 0.221  | 1          | 1.531                | 2.464                | 4.899%                   | 8.073%              |

Figure 22: Modal result summary

Results

Results

Export current tab content

Summary DOF Map Matrices Modal Table Mode Shapes

Matrix **K**

K<sub>X</sub>, K<sub>Y</sub>, and C<sub>F</sub> are the reduced/condensed dynamic matrices used for modal analysis.

| DOF  | DOF1            | DOF2            | DOF3            |
|------|-----------------|-----------------|-----------------|
| DOF1 | 1702.341981157  | -2477.372693767 | 962.885943644   |
| DOF2 | -2477.372693767 | 6147.807745223  | -5335.98749945  |
| DOF3 | 962.885943644   | -5335.98749945  | 10281.760500691 |

Figure 23: Modal stiffness matrix

Results

Results

Export current tab content

Summary DOF Map Matrices Modal Table Mode Shapes

Matrix **M**

K<sub>X</sub>, K<sub>Y</sub>, and C<sub>F</sub> are the reduced/condensed dynamic matrices used for modal analysis.

| DOF  | DOF1 | DOF2 | DOF3 |
|------|------|------|------|
| DOF1 | 15   | 0    | 0    |
| DOF2 | 0    | 15   | 0    |
| DOF3 | 0    | 0    | 20   |

Figure 24: Modal mass matrix

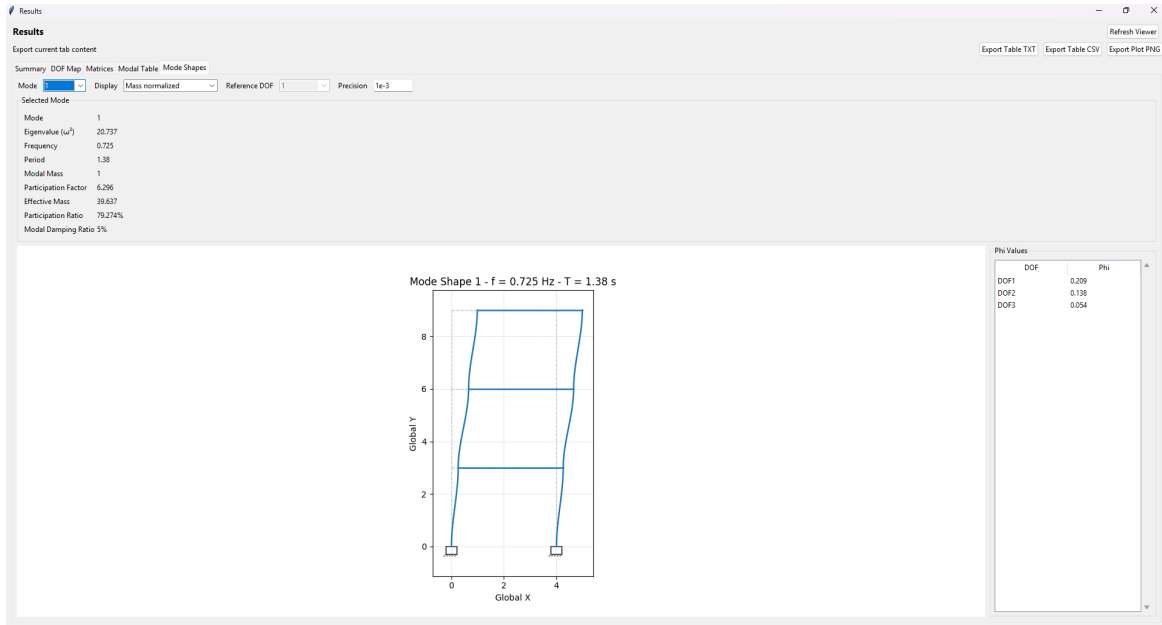


Figure 25: Mode-shape visualization

## 5 Static Frame Benchmark

The static frame benchmark verifies the direct stiffness workflow on a two-dimensional frame/truss-style model using retained XML input. The developed software path is:

XMLParser → DOFOptimizer → MatrixAssembler  
→ BandedSolver → PostProcessor → StaticResults

The SAP2000 output is parsed into the project coordinate convention before comparison. This benchmark checks that the global stiffness assembly, boundary conditions, displacement solution, reaction recovery, and member-end force recovery remain consistent with the reference solution.

Table 2: Static result table export evidence from `data/test_frame_static_results_table.txt`

| Exported field      | Exported value     |
|---------------------|--------------------|
| Model               | New Model          |
| Load case           | LC1                |
| Nodes and members   | 4 nodes; 4 members |
| Displacement rows   | 4                  |
| Reaction rows       | 2                  |
| Member force groups | 4                  |
| Units               | kN_m_tonne         |

The available static export is a visible-table summary, not a full numerical export of displacements, reactions, and member forces. It verifies the retained desktop export content and row counts. The benchmark comparison rows below still require a final exported numerical static table before the developed result and percent-difference cells can be completed.

Table 3: Static frame benchmark comparison table

| Quantity                                             | Developed software result | Reference result                                                                            | Percent difference | Tolerance           | Pass/fail |
|------------------------------------------------------|---------------------------|---------------------------------------------------------------------------------------------|--------------------|---------------------|-----------|
| Representative nodal displacement<br>$u_x, u_y, r_z$ | TODO                      | SAP2000 joint 2:<br>$U_1 = 0.031277$ m,<br>$U_3 = 0.000545$ m                               | TODO               | Benchmark tolerance | TODO      |
| Representative support reaction<br>$F_x, F_y, M_z$   | TODO                      | SAP2000 joint 1:<br>$F_1 = -20.000$ kN, $F_3 = -5.000$ kN                                   | TODO               | Benchmark tolerance | TODO      |
| Representative member-end force at I end             | TODO                      | SAP2000 frame F1 joint 1:<br>$F_1 = -14.514$ kN, $F_3 = -0.727$ kN,<br>$M_2 = -3.5389$ kN-m | TODO               | Benchmark tolerance | TODO      |
| Representative member-end force at J end             | TODO                      | SAP2000 frame F1 joint 2:<br>$F_1 = 14.514$ kN, $F_3 = 0.727$ kN,<br>$M_2 = -40.0018$ kN-m  | TODO               | Benchmark tolerance | TODO      |
| Selected N/V/M value or diagram endpoint             | TODO                      | TODO                                                                                        | TODO               | Engineering check   | TODO      |

### Static frame benchmark screenshot

TODO placeholder for SAP2000/reference output and developed software static result screenshot or exported table. Remaining TODO values require exported developed-software numerical rows, reference rows, or screenshots that are not present in the available evidence files.

*TODO: Insert final screenshot or exported comparison table.*

Figure 26: Static frame benchmark screenshot



## 6 Settlement Benchmark

The settlement benchmark verifies prescribed support displacement handling. For restrained displacement vector  $u_r$ , the reduced system solved by the static workflow is

$$K_{ff}u_f = F_f - K_{fr}u_r.$$

After the full displacement vector is reconstructed, reactions are recovered from  $R = Ku - F$ . The retained benchmark includes a vertical settlement at a restrained support and compares the resulting displacement and reaction fields with SAP2000 output.

Table 4: Settlement benchmark comparison table

| Quantity                                     | Developed software result    | Reference result                                                     | Percent difference | Tolerance                              | Pass/fail |
|----------------------------------------------|------------------------------|----------------------------------------------------------------------|--------------------|----------------------------------------|-----------|
| Prescribed settlement value used in assembly | XML node 5: $u_y = -0.002$ m | Same XML input                                                       | 0.000%             | Exact input check                      | Pass      |
| Representative free nodal displacement       | TODO                         | SAP2000 joint 2: $U_1 = 3.549 \times 10^{-6}$ m, $U_3 = -0.001994$ m | TODO               | SAP2000 comparison tolerance           | TODO      |
| Representative settled-support reaction      | TODO                         | SAP2000 joint 5: $F_1 = 4.993$ kN, $F_3 = -11.637$ kN                | TODO               | SAP2000 comparison tolerance           | TODO      |
| Representative member-end force              | TODO                         | SAP2000 frame F4 joint 5: $F_1 = 4.993$ kN, $F_3 = -11.637$ kN       | TODO               | SAP2000 comparison tolerance           | TODO      |
| SAP2000 display-mapping row                  | TODO                         | SAP2000-display node 5: $U_3 = -0.002000$ m, $R_2 = -0.000082$ rad   | TODO               | $5.0 \times 10^{-6}$ display tolerance | TODO      |

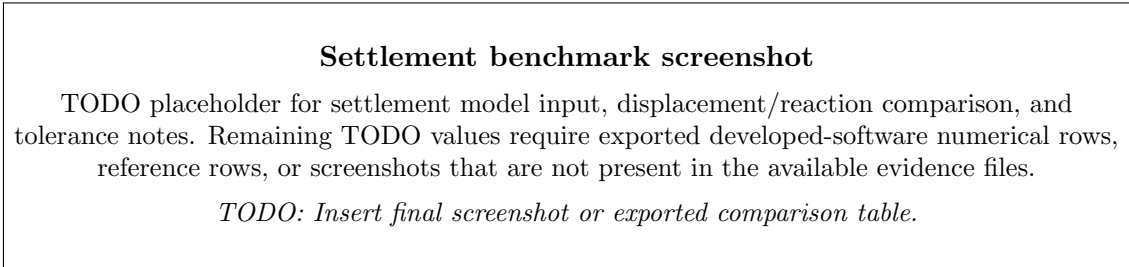


Figure 27: Settlement benchmark screenshot

## 7 Temperature Benchmark

The temperature benchmark verifies thermal fixed-end effects and transformation-sensitive force recovery. The XML input contains uniform and combined temperature loads on frame and truss

members. The inclined truss members are especially useful because errors in local/global transformation, thermal force sign, or SAP2000 coordinate mapping create visible force mismatches.

The strict thermal benchmark checks displacements, reactions, and member-end force components. The automated strict test uses a displacement tolerance of  $1.0 \times 10^{-5}$ , a force tolerance of 0.05, and a moment tolerance of 0.6 in the project units used by the benchmark.

Table 5: Temperature benchmark comparison table

| Quantity                               | Developed software result | Reference result                                                                             | Percent difference | Tolerance            | Pass/fail |
|----------------------------------------|---------------------------|----------------------------------------------------------------------------------------------|--------------------|----------------------|-----------|
| Representative thermal displacement    | TODO                      | SAP2000 joint 2:<br>$U_1 = -7.321 \times 10^{-7} \text{ m}$ ,<br>$U_3 = -0.001930 \text{ m}$ | TODO               | $1.0 \times 10^{-5}$ | TODO      |
| Representative support force reaction  | TODO                      | SAP2000 joint 1:<br>$F_1 = 1921.004 \text{ kN}$ , $F_3 = 5.911 \text{ kN}$                   | TODO               | 0.05                 | TODO      |
| Representative support moment reaction | TODO                      | SAP2000 joint 1:<br>$M_2 = -309.8603 \text{ kN-m}$                                           | TODO               | 0.6                  | TODO      |
| Inclined truss axial force             | TODO                      | SAP2000 truss T2 joint 5:<br>$F_1 = -8.247 \text{ kN}$                                       | TODO               | 0.05                 | TODO      |
| Frame member thermal end moment        | TODO                      | SAP2000 frame F1 joint 1:<br>$M_2 = -309.8603 \text{ kN-m}$                                  | TODO               | 0.6                  | TODO      |

|                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p style="text-align: center;"><b>Temperature benchmark screenshot</b></p> <p>TODO placeholder for temperature-load comparison output. Remaining TODO values require exported developed-software numerical rows, reference rows, or screenshots that are not present in the available evidence files.</p> <p style="text-align: center;"><i>TODO: Insert final screenshot or exported comparison table.</i></p> |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Figure 28: Temperature benchmark screenshot

## 8 Modal Flexible-Beam / Shear-Frame Benchmark

The modal benchmark verifies that the dynamic workflow preserves stiffness coupling while avoiding invalid massless dynamic DOFs. The benchmark uses a flexible-beam/shear-frame-style model, assembles the stiffness and lumped mass matrices, condenses stiffness-coupled massless DOFs where required, and solves

$$K\phi = \lambda M\phi.$$

The comparison checks the reduced or condensed stiffness matrix, eigenvalues, circular frequencies, periods, and normalized mode shapes against the stored reference solution.

Table 6: Modal result export values from `data/modal_results_table.csv`

| Mode | $\lambda$ | $\omega$ | $f$   | $T$   | Modal mass | $\Gamma$ | Effective mass | Mass ratio |
|------|-----------|----------|-------|-------|------------|----------|----------------|------------|
| 1    | 20.737    | 4.554    | 0.725 | 1.380 | 1.000      | 6.296    | 39.637         | 79.274%    |
| 2    | 209.652   | 14.479   | 2.304 | 0.434 | 1.000      | 2.821    | 7.958          | 15.917%    |
| 3    | 807.042   | 28.408   | 4.521 | 0.221 | 1.000      | 1.551    | 2.404          | 4.809%     |

Table 7: Modal flexible-beam benchmark comparison table

| Quantity                            | Developed software result | Reference result                    | Percent difference | Tolerance            | Pass/fail |
|-------------------------------------|---------------------------|-------------------------------------|--------------------|----------------------|-----------|
| Condensed/reduced $K_c(1, 1)$ entry | TODO                      | $1.0282 \times 10^4$ kN/m           | TODO               | 1.0 absolute         | TODO      |
| Eigenvalue $\lambda_1$              | 20.737                    | 20.739                              | 0.010%             | 0.02 absolute        | Pass      |
| Circular frequency $\omega_1$       | 4.554                     | 4.554                               | 0.000%             | 0.002 absolute       | Pass      |
| Period $T_1$                        | 1.380                     | 1.380                               | 0.000%             | 0.002 absolute       | Pass      |
| Normalized mode-shape component     | TODO                      | Top-story mode 1 component = 1.0000 | TODO               | $1.0 \times 10^{-3}$ | TODO      |
| Modal mass participation            | 79.274% for mode 1        | TODO                                | TODO               | Modal property check | TODO      |

### Mode shape plot

TODO placeholder for developed software mode-shape plot and reference mode-shape or frequency table. Remaining TODO values require exported developed-software numerical rows, reference rows, or screenshots that are not present in the available evidence files.

*TODO: Insert final screenshot or exported comparison table.*

Figure 29: Mode shape plot

## 9 Automated Pytest Validation

The final stable testing baseline is:

```
pytest tests/ -q -> 197 passed
pytest tests/test_ui_desktop_static.py -q -> 63 passed
pytest tests/test_ui_desktop_static.py
      tests/test_visualizer.py tests/test_visualizer_integration.py -q
      -> 76 passed
```

The final hardening changed only `tests/test_ui_desktop_static.py`. No solver math, XML schema, fixtures, or source behavior changed during final testing hardening. Static benchmark tests now use retained `data/test-frame.xml`. The file `data/ground_motion.txt` was restored only for legacy/future THA backend compatibility and does not change the final desktop scope.

```
(.venv) PS C:\Users\MOHAMMAD UMAIR\Documents\GitHub\CE4811-Final-Project-V2> pytest tests/
platform win32 -- Python 3.14.3, pytest-9.0.3, pluggy-1.6.0
rootdir: C:\Users\MOHAMMAD UMAIR\Documents\GitHub\CE4811-Final-Project-V2
plugins: anyio-4.13.0
collected 197 items

tests/test_controller.py ..... [ 25%]
tests/test_dynamic_assembly.py ..... [ 95%]
tests/test_dynamics.py ... [ 105%]
tests/test_interface.py ..... [ 125%]
tests/test_modal_solver.py ... [ 145%]
tests/test_model.py ..... [ 165%]
tests/test_model_builder.py ..... [ 205%]
tests/test_regression.py ..... [ 225%]
tests/test_response_spectrum.py ..... [ 245%]
tests/test_sap2000_strict.py ..... [ 315%]
tests/test_static.py ..... [ 345%]
tests/test_time_history.py ..... [ 395%]
tests/test_ui_desktop_static.py ..... [ 425%]
tests/test_ui_dynamic.py ..... [ 445%]
tests/test_ui_model_input.py ..... [ 475%]
tests/test_ui_results.py .. [ 485%]
tests/test_ui_shell.py ..... [ 515%]
tests/test_ui_static.py ..... [ 545%]
tests/test_ui_visualization.py . [ 555%]
tests/test_unit.py ..... [ 595%]
tests/test_visualizer.py ..... [ 655%]
tests/test_visualizer_integration.py ..... [ 685%]

----- 197 passed in 2.19s -----
(.venv) PS C:\Users\MOHAMMAD UMAIR\Documents\GitHub\CE4811-Final-Project-V2>
```

Figure 30: Test suite output

Table 8: Automated test coverage summary

| Test area                           | Representative tests                                      | Checked behavior                                                                                                 | Result |
|-------------------------------------|-----------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|--------|
| XML parsing/export                  | XML parser/export tests and desktop XML command-path test | Save/load/export reproducibility and parse-back through XMLParser                                                | Passed |
| Model building                      | ModelBuilder tests and template workflows                 | Programmatic, template, table, and XML-backed model creation                                                     | Passed |
| DOF and stiffness assembly          | Static solver and assembly tests                          | DOF maps, element stiffness, global $K$ , reduced $K_{ff}$ , and force vectors                                   | Passed |
| Static result recovery              | Static, post-processing, and benchmark tests              | Displacements, reactions, member forces, member-end force signs, and N/V/M diagram data                          | Passed |
| Settlement and temperature behavior | Interface, unit, strict SAP2000, and regression tests     | Prescribed support displacement assembly, thermal fixed-end effects, and transformation-sensitive force recovery | Passed |
| Dynamic/modal assembly              | Modal solver, dynamic assembly, and regression tests      | Mass assembly, active dynamic DOFs, massless-DOF handling, eigenvalues, frequencies, and periods                 | Passed |

| Test area               | Representative tests                        | Checked behavior                                                                                                                                                      | Result |
|-------------------------|---------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| Modal properties        | Modal solver and desktop modal tests        | Mass normalization, participation factors, effective modal mass, participation ratios, and display normalization options                                              | Passed |
| Visualization           | Visualizer and visualizer integration tests | Model preview, deformed shape, diagrams, mode-shape plotting adapters, and missing-field tolerance                                                                    | Passed |
| Desktop workflow/export | UI desktop static tests                     | Static table TXT export, Modal table CSV export, Static plot PNG export, XML save/export parse-back, stale-result clearing, and final Static/Modal-only desktop scope | Passed |
| Reference validation    | SAP2000/reference comparison tests          | Agreement with stored reference solutions within the tolerances described above                                                                                       | Passed |

## 10 Desktop XML/Export Command-Path Validation

Two final P0 desktop workflow tests were added during final hardening:

1. A headless XML command-path test for `_save_xml()` and `_export_xml()`, with the written XML parsed back through `XMLParser`.
2. A result export test for Static table TXT export, Modal table CSV export, and Static plot PNG export through the shared export path.

These tests address the previous main risk area: desktop workflow coverage rather than numerical correctness. They confirm that the final UI command paths reach the XML and result-export backends without requiring a manual GUI session.

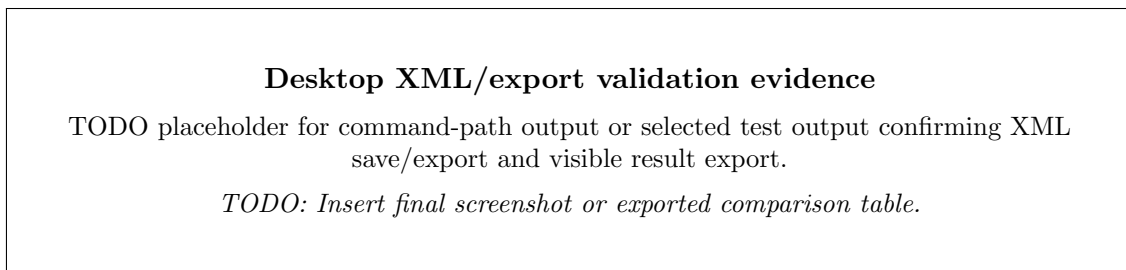


Figure 31: Desktop XML/export validation evidence

## 11 Remaining Verification Inserts

Before final submission, replace the remaining placeholder comparison cells with final values from the selected static frame, settlement, temperature, and modal benchmark exports. Each comparison table should preserve the required columns: Quantity, Developed software result, Reference result, Percent difference, Tolerance, and Pass/fail.

Remaining TODO placeholders in this appendix are:

Table 9: Desktop export evidence from retained **data/** outputs

| Export artifact | Export type         | Verified evidence in retained file                                                                                                                                                                                                          |
|-----------------|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Static table    | Static TXT table    | Source <code>data/test_frame_static_results_table.txt</code> : model <code>New Model</code> ; load case <code>LC1</code> ; 4 displacement rows; 2 reaction rows; 4 member-force groups; <code>kN_m.tonne</code> units.                      |
| Static plots    | PNG figure evidence | Static plot figures are already placed under <code>docs/final_latex/main_report/figures/</code> : deformed shape, axial force, shear force, and bending moment. No standalone static plot export file is present under <code>data/</code> . |
| Modal table     | Modal CSV table     | Source <code>data/modal_results_table.csv</code> : three modal rows with eigenvalues, frequencies, periods, modal mass, participation factors, effective modal mass, participation ratios, and damping ratios.                              |
| Modal plots     | Modal PNG plots     | Source <code>data/modal_results_plot*.png</code> : three exported modal plot PNG files exist in <code>data/</code> ; LaTeX figure references remain limited to images already placed under <code>docs/final_latex/</code> .                 |

- Static frame developed-software numerical comparison rows, N/V/M value, and screenshot evidence.
- Settlement developed-software numerical comparison rows and screenshot evidence.
- Temperature developed-software numerical comparison rows and screenshot evidence.
- Modal condensed matrix developed value, mode-shape developed value, modal participation reference value, and mode-shape evidence.
- Desktop XML/export command-path evidence screenshot.